1.

(a)

$x \backslash yz$	00	01	11	10
0	1	0	1	1
1	0	0	1	0

yz $x'z'$

$x'z' + yz$ #

(b)

$x \backslash yz$	00	01	11	10
0	0	0	1	1
1	0	1	1	1

y xz

$xz + y$ #

(c)

$x \backslash yz$	00	01	11	10
0	1	1	0	1
1	1	0	0	1

$x'y'$ z'

$z' + x'y'$ #

(d)

$x \backslash yz$	00	01	11	10
0	0	1	1	0
1	0	1	1	0

z

z #

2.

(a)

$x \backslash yz$	00	01	11	10
0	0	0	1	1
1	1	1	1	0

$x'y$ xy' yz

$xy' + x'y + yz$ #

(b)

$x \backslash yz$	00	01	11	10
0	1	1	0	0
1	1	1	1	1

y' x

$x + y'$ #

(c)

$x \backslash yz$	00	01	11	10
0	1	1	0	1
1	1	1	0	1

z' y'

$y' + z'$ #

(d)

$x \backslash yz$	00	01	11	10
0	0	1	1	1
1	0	1	1	1

y z

$y + z$ #

(e)

$x \backslash yz$	00	01	11	10
0	0	1	1	0
1	0	1	1	0

z

z #

(f)

$x \backslash yz$	00	01	11	10
0	1	1	1	1
1	0	1	1	0

x' z

$x'z$ #

3.

(a) $x'y'z + xyz' + x'yz + xy$

001, 110, 011 11x

x\yz	00	01	11	10
0	0	1	1	0
1	0	0	1	1

$x'z$, xy

$x'z + xy$ *

(b) $xyz' + y'z' + xz$

110, x00, 1x1

x\yz	00	01	11	10
0	1	0	0	0
1	1	1	1	1

$y'z'$, x

$x + y'z'$

(c) $x'yz + xy' + yz'$

011, 10x, x10

x\yz	00	01	11	10
0	0	0	1	1
1	1	1	0	1

$x'y$, xy', yz'

$xy' + x'y + yz'$

(d) $xz' + x'z + x'y + xy'$

1x0, 0x1, 01x, 10x

x\yz	00	01	11	10
0	0	1	1	1
1	1	1	0	1

$x'z$, xy', yz'

$xy' + x'z + yz'$

4.

(a)

wx\yz	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	0	0	0
10	1	0	0	0

$w'y'$, $y'z'$

$w'y' + y'z'$ *

(b)

wx\yz	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

xz', $wx'z$

$wx'z + xz'$

4.

(c)

wx \ yz	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	0	0	0	0
10	0	1	1	0

$w'z$ $\quad$ $w'xy$ $\quad$ $x'z$

$w'z+x'z+w'xy$

(d)

wx \ yz	00	01	11	10
00	0	0	0	1
01	0	0	1	1
11	0	0	1	1
10	0	0	0	1

yz' $\quad$ xy

$xy+yz'$

(e)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

wxy' $\quad$ $wx'y$ $\quad$ wz

$wz+wxy'+wx'y$ #

(f)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'z'$ $\quad$ wz

$wz+w'z'$ #

(g)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'z'$ $\quad$ $x'y'$

$x'y'+w'z'$ #

(h)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

xy $\quad$ $x'y'z'$ $\quad$ yz

$xy+yz+x'y'z'$

5.

(a)

$wx \backslash yz$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'x'y$, $w'z'$, $y'z'$, $wx'y'$

$w'z' + y'z' + wx'y' + w'x'y$ #

(b)

$wx \backslash yz$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'x'$, $x'y'z'$, $xy'z$

$x'y'z' + xy'z + w'x'$ #

(c)

$wx \backslash yz$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'x'y$, $w'yz$, $xy'z$, $x'yz$

$x'yz + xy'z + w'yz + w'x'y$

(d)

$wx \backslash yz$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'x'z$, $w'y$, wy'

$w'y + wy' + w'x'z$ #

6.

(a) $ABC'D' + AB'C + B'C'D'$

1100, 101x, x000

$+ AB'CD + B'C'D$

1011, x001

$AB \backslash CD$	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	1	0	0	0
10	1	1	1	1

$AC'D'$, $B'C'$, AB'

$AC'D' + B'C' + AB'$ #

(b) $w'x'y' + w'x'yz + x'y'z + xyz + y'z'$

000x, 0011, x001, x111, xx00

$w'x' + x'y' + xyz + w'yz$ #

$wx \backslash yz$	00	01	11	10
00	1	1	1	0
01	1	0	1	0
11	1	0	1	0
10	1	1	0	0

$w'yz$, xyz, $w'x'$, $x'y'$

6.

(c) $A'B'CD + A'BC + C'D$

0011, 011X, XX01,

$+ ABCD + AB'C$

1111, 101X

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	0	1	1	0
10	0	1	1	1

$A'BC$

$AB'C$

D

$A'BC + AB'C + D$ #

(d) $A'B'C' + A'BD + A'BC'D' + BC'D + ABCD$

000X, 01X1, 0100, X101, 1111

AB \ CD	00	01	11	10
00	1	1	0	0
01	1	1	1	0
11	0	1	1	0
10	0	0	0	0

$A'C'$

$B'D$

$A'C' + B'D$ #

7.

(a) $w'x'z + xy' + w'x + wxy$

00X1, X10X, 01XX, 111X

wx \ yz	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$w'z$

x

$x + w'z$ #

(b) $A'BD' + BDC + ABC' + BD + ABC$

01X0, X111, 110X, X1X1, 111X

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

B

B #

(c) $AB'C' + AC'D' + ABD + BC'D + A'BC'D'$

100X, 1X00, 11X1, X101, 0100

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	0	0
11	1	1	1	0
10	1	1	0	0

BC'

ABD

AC'

$AC' + BC' + ABD$ #

(d) $wxyz + wx' + wx'y + wxy + w'yz'$

1111, 10XX, 101X, 111X, 0X10

wx \ yz	00	01	11	10
00	0	0	0	1
01	0	0	0	1
11	0	0	1	1
10	1	1	1	1

yz'

wy

wx'

$wy + wx' + yz'$

8. find minterms.

(a) $w'x + xz + wyz$

01xx, x1x1, 1x11

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$\Sigma(4,5,6,7,11,13,15)$ #

(b) $ABD' + BC'D + CD$

11x0, x101, xx11

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$\Sigma(3,5,7,11,12,13,14,15)$ #

(c) $x'y'z' + xy + y'z$

000, 11x, x01

x \ yz	00	01	11	10
0	0	1	3	2
1	4	5	7	6

$\Sigma(0,1,5,6,7)$ #

(d) $AB + B'C'D' + BCD + A'C'D'$

11xx, x000, x111, 0x00

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$\Sigma(0,4,7,8,12,13,14,15)$

9. find PI (質隱項/質蘊項) & EPI (基本質隱項)

(a)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$x'z'$

xz

EPI：xz、$x'z'$ #

(b)

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$A'BD$

AC

$B'D'$

EPI：AC、$B'D'$、$A'BD$ #

9.

(c)

AB\CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$A'C$

BC'

$AB'D$

EPI: BC'、$AB'D$、$A'C$ #

(d)

wx\yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$w'x'z$

xy

wy'

EPI: wy'、xy、$w'x'z$ #

(e)

①

AB\CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$B'D'$

BD

EPI: BD, $B'D'$

(f)

wx\yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$x'z'$

xyz

EPI: $x'z'$, xyz

10.

(a)

wx\yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$x'z'$

wz'

xz

EPI: xz、$x'z'$

$xz+x'z'+wz'$ #

(b)

AB\CD	00	01	11	10
00	0	1	3	2
01	4	5	6	7
11	12	13	15	14
10	8	9	11	10

$A'B'C$

CD'

AC

$B'D'$

$A'BC'D$

EPI: $A'BC'D$、CD'、AC, $B'D'$

$AC+CD'+B'D'+A'B'C+A'BC'D$ #

10.

(c)

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

A'B'D

AC

BC'

EPI: BC'、AC

BC'+AC+A'B'D #

(d)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

w'y'

xy

wx'z

EPI: w'y、xy

w'y+xy+wx'z #

(e)

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

B'C'

A'CD

ABD

EPI: AB'D'、B'C'

B'C'+AB'D'+ABD+A'CD #

(f)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

w'z'

w'y'

xyz

x'yz

EPI: w'y'、xyz、x'yz

w'y'+w'z'+xyz+x'yz #

11.

wx \ yz	00	01	11	10
00	0	1	0	1
01	1	1	0	0
11	0	1	0	1
10	0	1	0	1

w'xy

wy'z'

x'y'z'

yz

F' = (yz)+(wy'z')+(x'y'z')+(w'xy)

(F')' = [(yz)+(wy'z')+(x'y'z')+(w'xy)]'

= (yz)'(wy'z')'(x'y'z')'(w'xy)'

= (y'+z')(w'+y+z)(x+y+z)(w+x'+y') #

12.

(a)

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	1	1	0
10	0	1	1	0

$A'B'C$

$A'BC'$ AD

$A'BC' + AD + A'B'C$ #

(b)

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	1	1	0	1
10	0	1	1	0

$A'CD$

BD'

$B'D$

$AC'D$

$B'D + BD' + AC'D + A'CD$ #

13

(a) $x'y'z' + yz' + xy$

000, x10, 11x

x \ yz	00	01	11	10
0	1	0	0	1
1	0	0	1	1

$x'z'$

xy

xy' $x'z$

$F' = (xy') + (x'z)$

$(F')' = [(xy') + (x'z)]'$

$= (xy')'(x'z)'$

$= (x'+y)(x+z')$

$F = \underbrace{xy + x'z'}_{SOP} = \underbrace{(x'+y)(x+z')}_{POS}$

(b) $A'B + A'B'C + CD$

01XX, 001X, XX11

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	1	1
11	0	0	1	0
10	0	0	1	0

$A'C$

$A'B$

CD

AD' AC' $B'C'$

$F' = (AD') + (AC') + (B'C')$

$(F')' = [(AD') + (AC') + (B'C')]'$

$= (AD')'(AC')'(B'C')'$

$= (A'+D)(A'+C)(B+C)$

$F = \underbrace{CD + A'B + A'C}_{SOP}$

$= \underbrace{(A'+D)(A'+C)(B+C)}_{POS}$

13.

(c) $A'B'D' + A'D + BCD + BC'D'$

$00X0,\ 0XX1,\ X111,\ X100$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	0
11	1	0	1	0
10	0	0	0	0

$A'B'$, ACD, BCD', AB', $BC'D'$, $AC'D$

$F' = (AB') + (AC'D) + (BCD')$

$(F')' = [(AB') + (AC'D) + (BCD')]'$

$= (AB')'(AC'D)'(BCD')'$

$= (A'+B)(A'+C+D')(B'+C'+D)$

$F = \underline{A'B' + ACD + BC'D'}$ SOP

$= \underline{(A'+B)(A'+C+D')(B'+C'+D)}$ POS

(d) $A'B'C + A'C'D + A'BC + ABD$

$001X,\ 0X01,\ 011X,\ 11X1$

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	1	1	1
11	1	0	1	0
10	1	0	0	0

$A'C'D$, ACD, $A'BC$, AC, $B'C$, $AC'D'$, $A'C'D'$, $AC'D$

$F' = AC + B'C + A'C'D' + AC'D$

$(F')' = [(AC) + (B'C) + (A'C'D') + (AC'D)]'$

$= (AC)'(B'C)'(A'C'D')'(AC'D)'$

$= (A'+C')(B+C')(A+C+D)(A'+C+D')$

$F = ACD + A'BC + AC'D'$

$= (A'+C')(B+C')(A+C+D)(A'+C+D')$ #

14.

$A'B'C' + A'CD + A'C'D' + A'BD + ABC + AB'C$

$000X,\ 0X11,\ 0X00,\ 01X1,\ 111X,\ 101X$

1st way.

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	0	0	1	1
10	0	0	1	1

$A'C'$, $\rightarrow AC$, CD

$AC + A'C' + CD$ #

2nd way

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	0	0	1	1
10	0	0	1	1

$A'CD'$, AC'

$F' = AC' + A'CD'$

$(F')' = [(AC') + (A'CD')]'$

$= (AC')'(A'CD')' = (A'+C)(A+C'+D)$ #

3rd way

$F = AC + A'C' + CD$

$= C(A+D) + A'C'$ #

15.

(a)

$x \backslash yz$	00	01	11	10
0	0	~~1~~	3	2
1	4	5	~~7~~	~~6~~

$x'y'$ z

$F = x'y' + z$

$= \Sigma(0,1,3,5,7)$ #

(b)

$AB \backslash CD$	00	01	11	10
00	~~0~~	1	~~3~~	2
01	4	~~5~~	~~7~~	6
11	12	13	~~15~~	14
10	8	9	~~11~~	10

A' $BC'D'$

$F = A' + BC'D'$

$= \Sigma(0,1,2,3,4,5,6,7,12)$ #

(c)

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	~~4~~	5	~~7~~	6
11	~~12~~	13	~~15~~	14
10	8	~~9~~	11	10

$A'B$ CD

$F = A'B + CD$

$= \Sigma(3,4,5,6,7,11,15)$ #

(d)

$AB \backslash CD$	00	01	11	10
00	0	1	~~3~~	~~2~~
01	4	5	7	6
11	~~12~~	~~13~~	15	14
10	8	9	11	10

BC' AB AC'

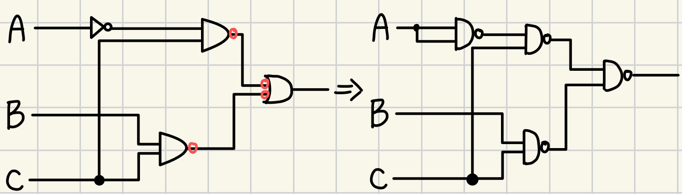
$F = AC' + BC' + AB$

$= \Sigma(4,5,8,9,12,13,14,15)$ #

16.

(a) $F(A,B,C,D) = A'B'C + A'BC + ABC$

001X, 011X, 111X



$AB \backslash CD$	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	0	0

$A'C$ BC

$F = A'C + BC$

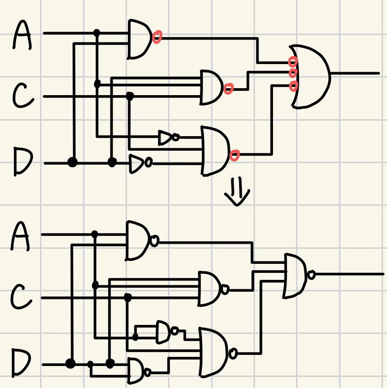
(b) $F(A,B,C,D) = A'CD' + A'BD + ABD + AB'CD$

0X10, 01X1, 11X1, 1011

$AB \backslash CD$	00	01	11	10
00	0	0	0	1
01	0	1	1	1
11	0	1	1	0
10	0	0	1	0

$A'CD'$ AD ACD

$F = AD + ACD + A'CD'$

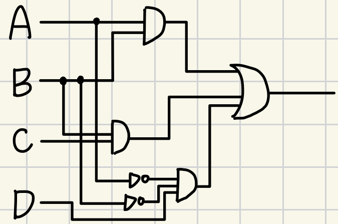


17.

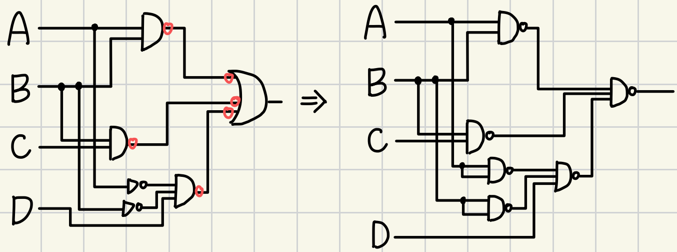
AB \ CD	00	01	11	10
00	~~0~~	1	3	~~2~~
01	~~4~~	~~5~~	7	6
11	12	13	15	14
10	~~8~~	~~9~~	~~11~~	~~10~~

A'B'D
BC
AB

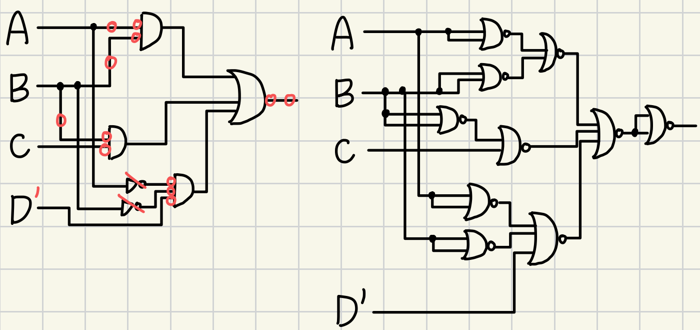
$F' = AB + BC + A'B'D$



(a)



(b)



19.

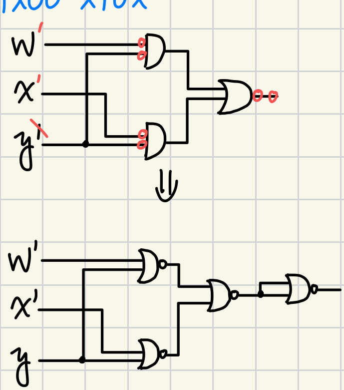
(a) $F(w,x,y,z) = wx'y' + wy'z' + xy'$

1 0 0 X   1 X 0 0   X 1 0 X

wx \ yz	00	01	11	10
00	0	0	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

wy'   xy'

$wy' + xy'$ ※

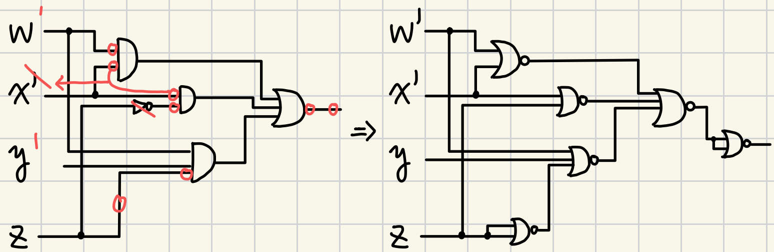


19.

(b)

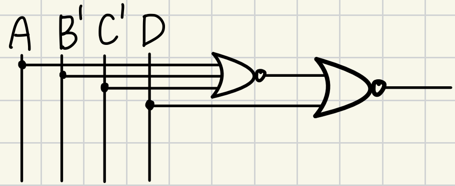
wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$F = wx' + x'z' + wyz$ ※



23.

AB \ CD	00	01	11	10
00	X	X	0	1
01	1	X	0	0
11	1	0	0	1
10	X	0	0	1



$F' = D + A'BC$

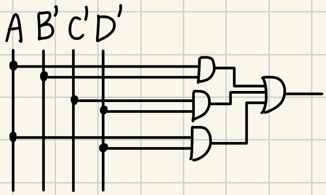
$F = [(D) + (A'BC)]'$

$= \{D + [(A + B' + C')]'\}'$

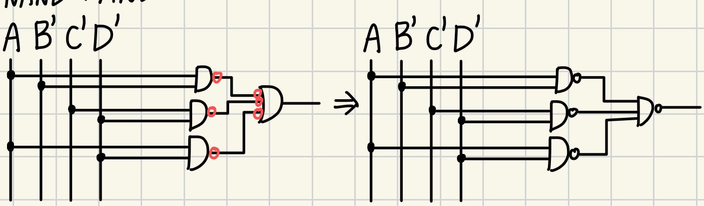
24.

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$F = AB' + C'D' + AD'$

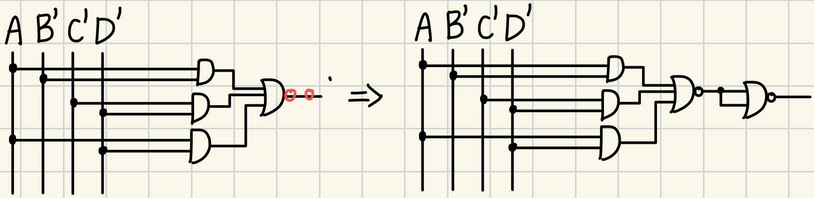


(a) NAND、AND

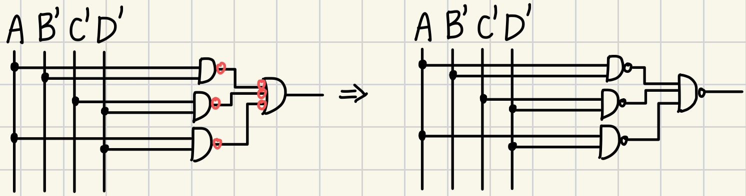


24.

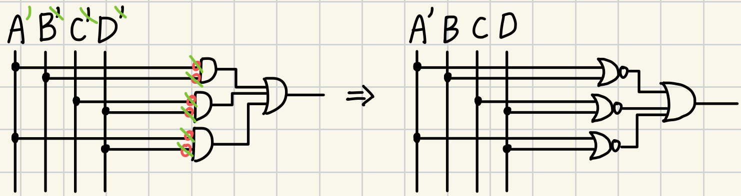
(b) AND、NOR



(c) OR 、NAND



(d) NOR、OR



26. $F = f \cdot g$, Find F.

$f = abc' + c'd + a'cd' + b'cd'$

110X, XX01, 0X10, X010

$g = (a+b+c'+d')(b'+c'+d)(a'+c+d')$

$g' = [(a+b+c'+d')(b'+c'+d)(a'+c+d')]'$

$= (a+b+c'+d')' + (b'+c'+d)' + (a'+c+d')'$

$= a'b'cd + bcd' + ac'd$ #

0011, X110, 1X01

f:

ab \ cd	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	1	1	0	0
10	0	1	0	1

g:

ab \ cd	00	01	11	10
00	1	1	0	1
01	1	1	1	0
11	1	0	1	1
10	1	0	1	1

$F = abc'd' + a'c'd + b'cd'$ #

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$F = AB'CD' + A'BCD' + AB'C'D + A'BC'D$

$= (AB' + A'B)CD' + (AB' + A'B)C'D$

$= (AB' + A'B)(CD' + C'D)$

$= (A \oplus B)(C \oplus D)$ ※